22. The name 'Blue John' is given to which of the following compounds ?

a) CaH_2 b) CaF_2 c) $Ca_3(PO_4)_2$ d) CaO

23. Formula of Gypsum is

a) $CaSO_4 . 2H_2O$ b) $CaSO_4 . ½ H_2O$

c) $3 CaSO_4 . H_2O$ d) $2CaSO_4 . 2H_2O$

24. When CaC_2 is heated in atmospheric nitrogen in an electric furnace the compound formed is

a) $Ca(CN)_2$ b) CaNCN

c) CaC_2N_2 d) $CaNC_2$

25. Among the following the least thermally stable is

(a) K_2CO_3 b) Na_2CO_3

(c) $BaCo_3$ d) Li_2CO_3

II. Write brief answer to the following questions:

26. Why sodium hydroxide is much more water soluble than sodium chloride?

27. Write the chemical equations for the reactions involved in solvay process of preparation of sodium carbonate.

28. An alkali metal (x) forms a hydrated sulphate, $X_2SO_4. 10H_2O$. Is the metal more likely to be sodium (or) potassium.

29. Write balanced chemical equation for each of the following chemical reactions.

(i) Lithium metal with nitrogen gas

(ii) heating solid sodium bicarbonate

(iii) Rubidum with oxygen gas

(iv) solid potassium hydroxide with CO_2

(v) heating calcium carbonate

(vi) heating calcium with oxygen

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